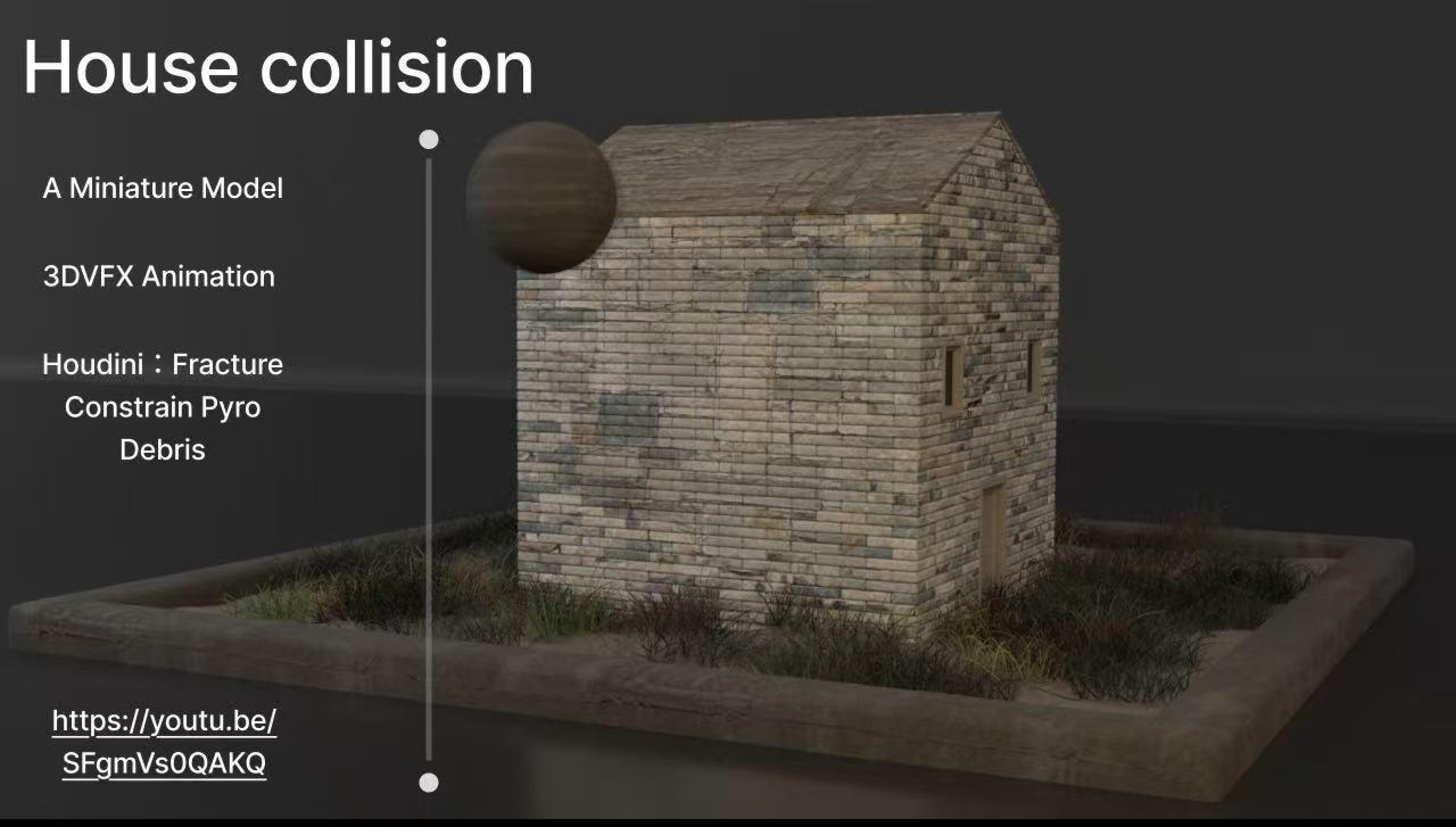
House collision



A Miniature Model

3DVFX Animation

Houdini：Fracture
Constrain Pyro
Debris

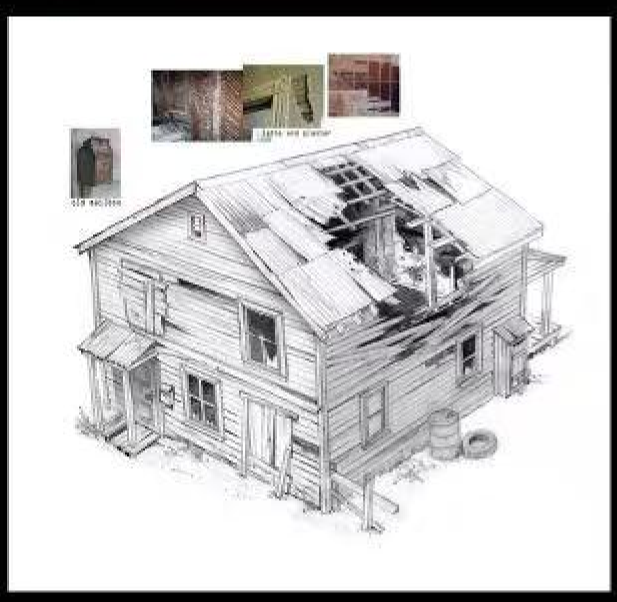
https://youtu.be/SFgmVs0QAKQ

《Love Death Robot》
-Night of mini dead-

Content
“Night of the Mini Dead” is a short episode from Love, Death & Robots. It tells the story of a zombie outbreak shown in a miniature, toy-like world. The episode begins with a funny accident in a cemetery, which brings the dead back to life. The tiny zombies spread quickly across cities and countries, causing chaos and destruction everywhere. Even the military and nuclear weapons cannot stop them. In the end, the entire Earth is destroyed, shown in a darkly humorous and playful style.

Inspiration

Reference

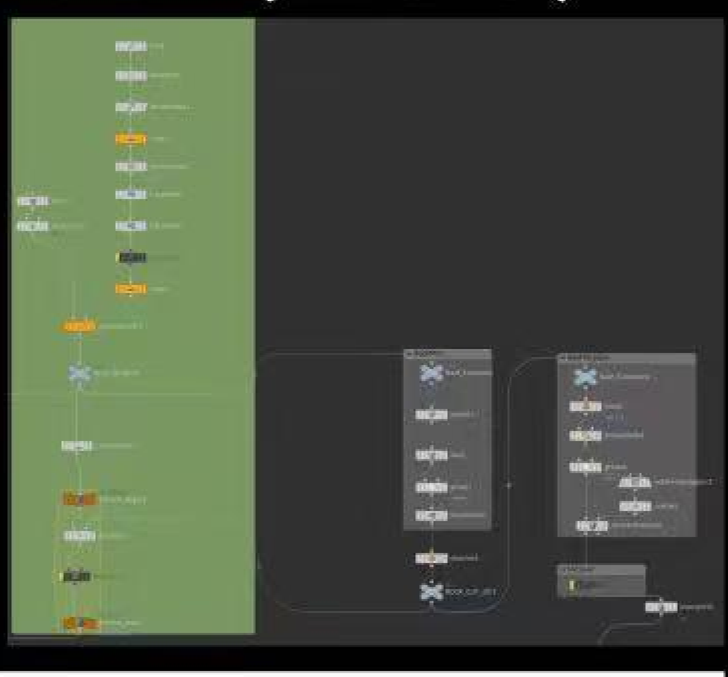


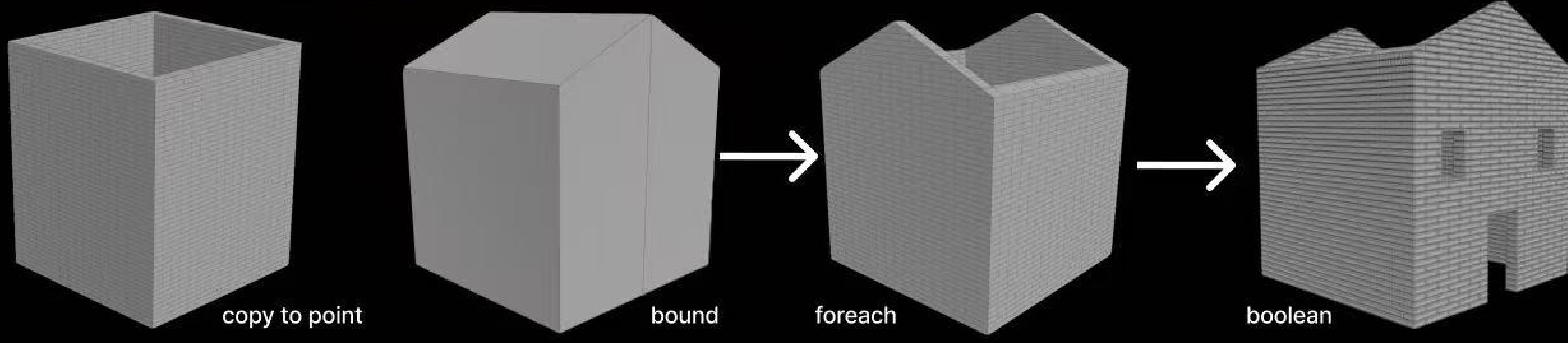
Process（PCG）

The PCG stage generates masks or cut-out shapes to control where and how fracturing occurs.

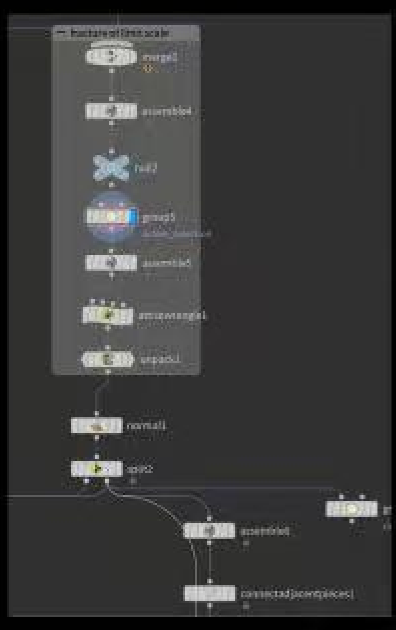
- Roof PCG
  - bound + box + group + transform + dissolve → creates bounding regions to limit where the roof will be fractured.
    - Why important? Instead of fracturing the entire roof randomly, this defines targeted breaking zones (e.g., center collapse).
- Door_Window PCG connectivity + foreach + boolean + polybevel
  - Similar, but applied to doors and windows. Ensures their geometry is pre-cut for destruction.

👉 Summary: PCG ensures art-directed destruction. Instead of uncontrolled random breaks, the artist can decide where the building should collapse first.

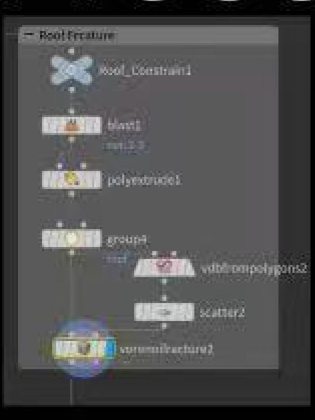




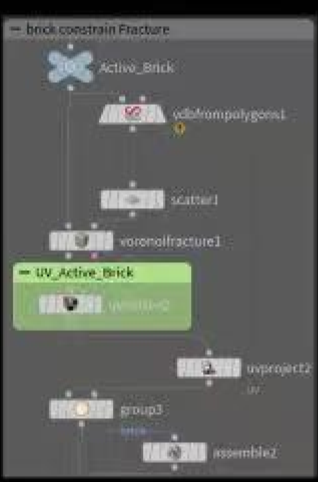
Process （Fracture）

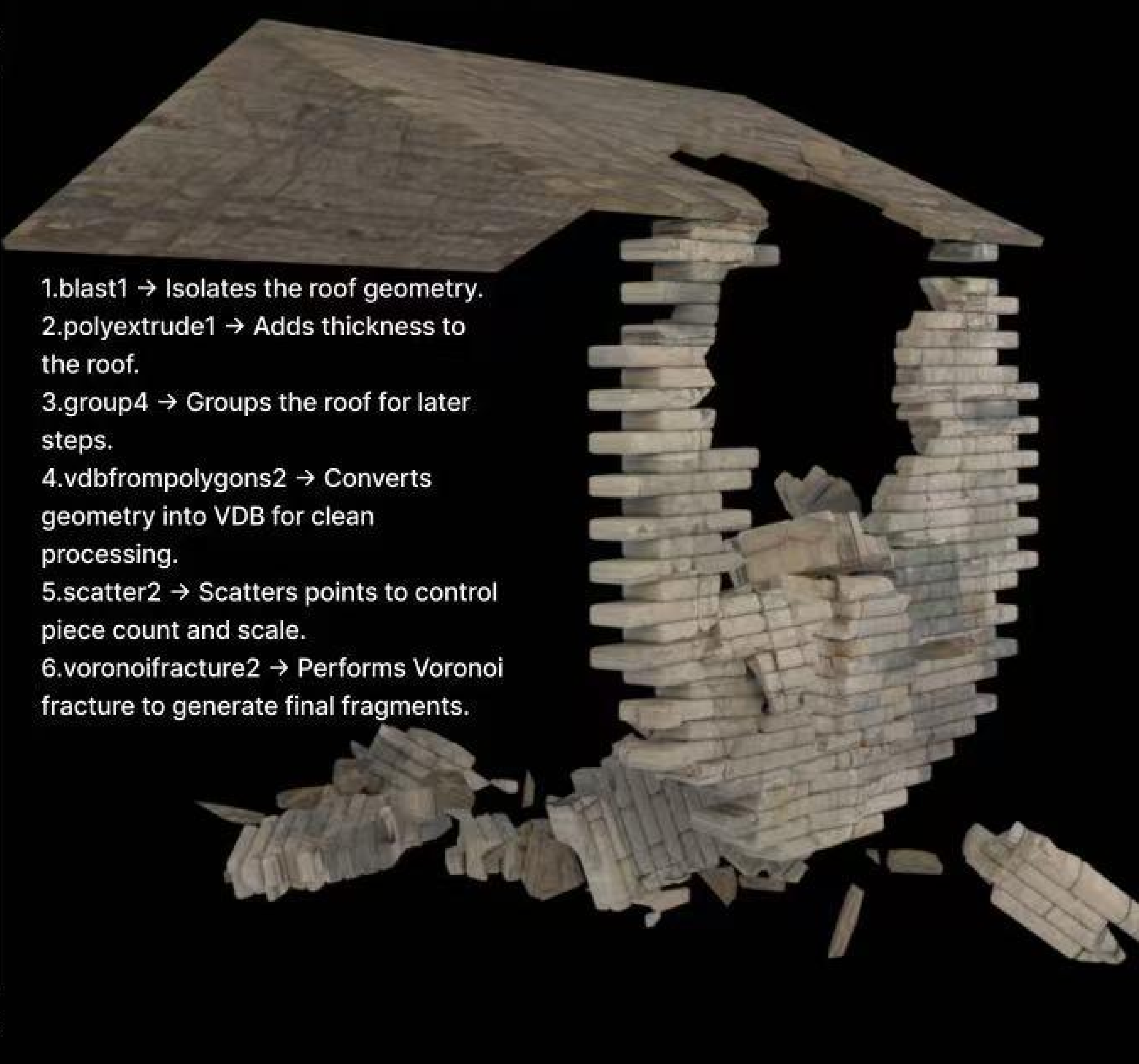


1. Merge input geometries.
2. Assign name attributes to fragments.
3. Use group5 to select a specific region → this is the key step of limiting the fracture range.
4. Reassemble and assign attributes (active/scale).
5. Unpack pieces for further operations (normals, splitting, etc.).



1. blast1 → Isolates the roof geometry.
2. polyextrude1 → Adds thickness to the roof.
3. group4 → Groups the roof for later steps.
4. vdbfrompolygons2 → Converts geometry into VDB for clean processing.
5. scatter2 → Scatters points to control piece count and scale.
6. voronoifracture2 → Performs Voronoi fracture to generate final fragments.





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